UE26CY151A Engineering Chemistry · Unit 2 · Electrochemistry and Corrosion

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 003 · Types of electrodes · compiled 17 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S003·5	Deck 003, slide 5 (rendered .pptx)
T003 00:07:50	AV-summary transcript 003, segment start
PDF p.57	Older consolidated Unit 2 slide PDF (8enterprise), page 57
ESA Dec-23 2d	Past paper: ESA Dec 2023, question 2.d
ESA Jan-25 s1 2b	Past paper: ESA Jan 2025 set 1, question 2.b

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Every figure is redrawn as a schematic (not to scale) with the Okabe–Ito colours: blue for solutions and reduced species, vermilion for the oxidised form and charge, orange for gases. G6.2 (potential against pH) is added for clarity; the glass line uses an illustrative E°_{G}, since E°_{G} differs from electrode to electrode.

2 Metal–metal ion electrode

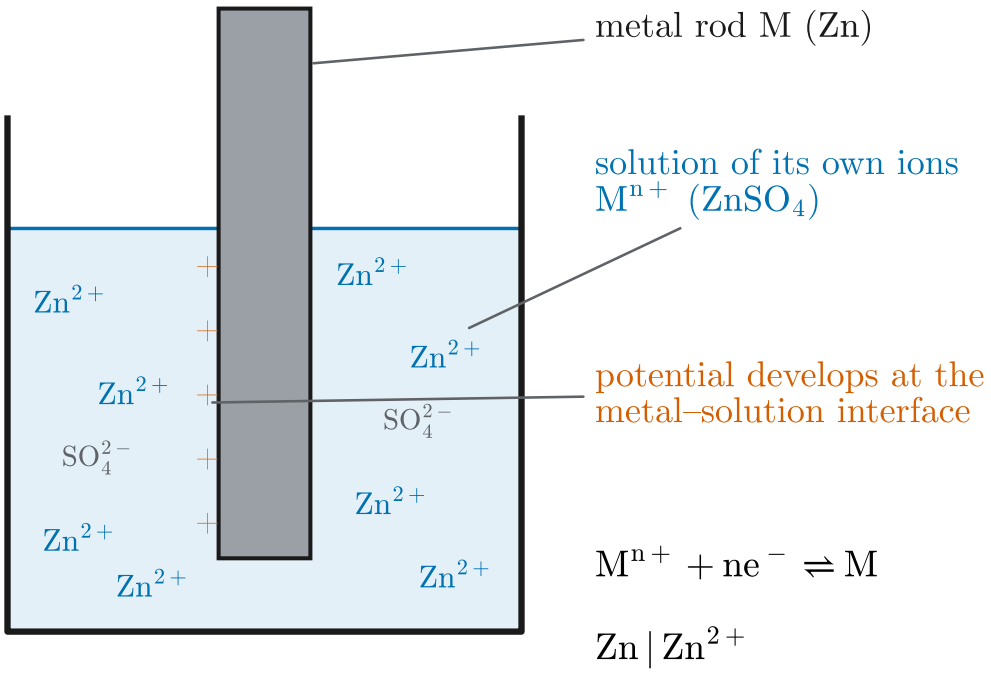


Figure G2.1. Metal–metal ion electrode ($\mathbf{Zn|Zn^{2+}}$). ESA ★

S003·3 · T003 00:01:26

How to read it. A zinc rod dips in a solution of its own ions. The potential develops at the rod–solution interface (marked +). Label: metal rod, solution of its own ions, the interface; write the reaction and the representation under it. Schematic, not to scale.

3 Metal–metal insoluble salt–anion electrode

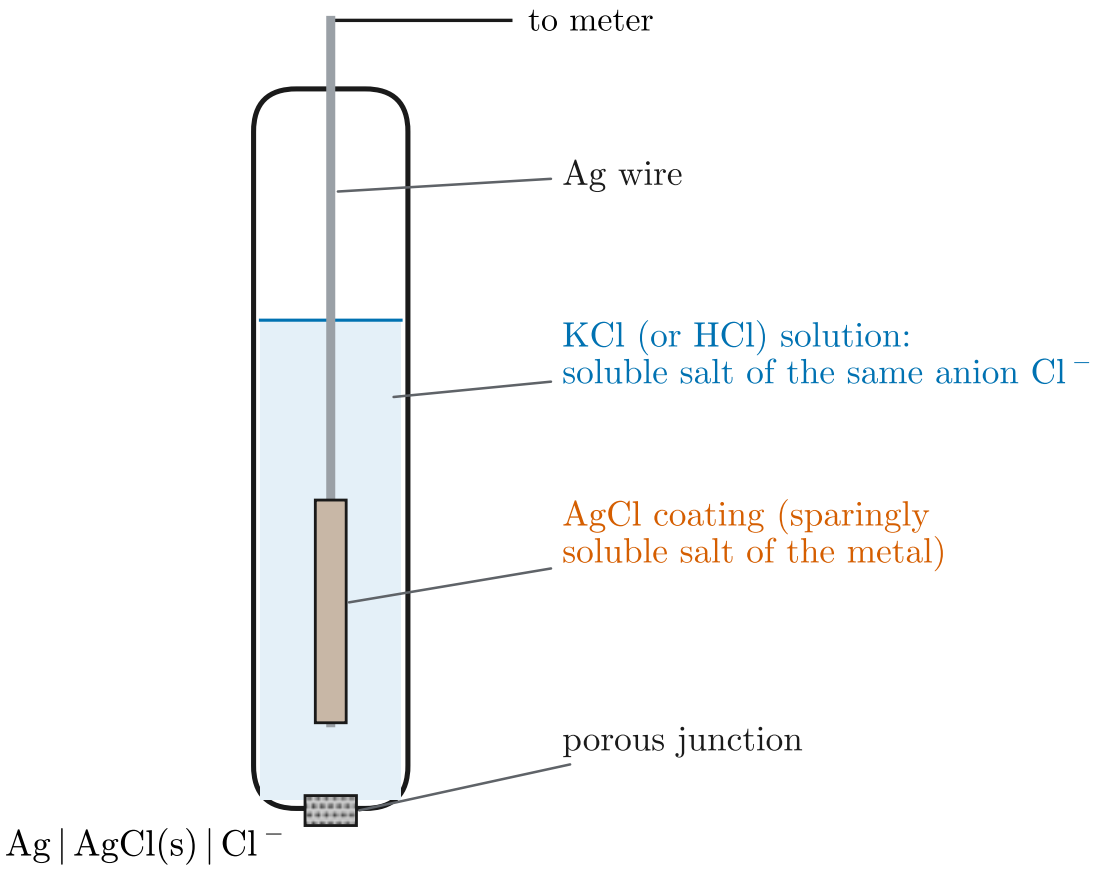


Figure G3.1. Silver–silver chloride electrode. ESA ★

S003·4 · T003 00:04:34

How to read it. A silver wire, its lower end coated with AgCl, sits in KCl (or HCl) inside a glass tube that meets the outer solution through a porous junction. The three labels to remember: metal, sparingly soluble salt of the metal, soluble salt of the same anion.

4 Gas electrode

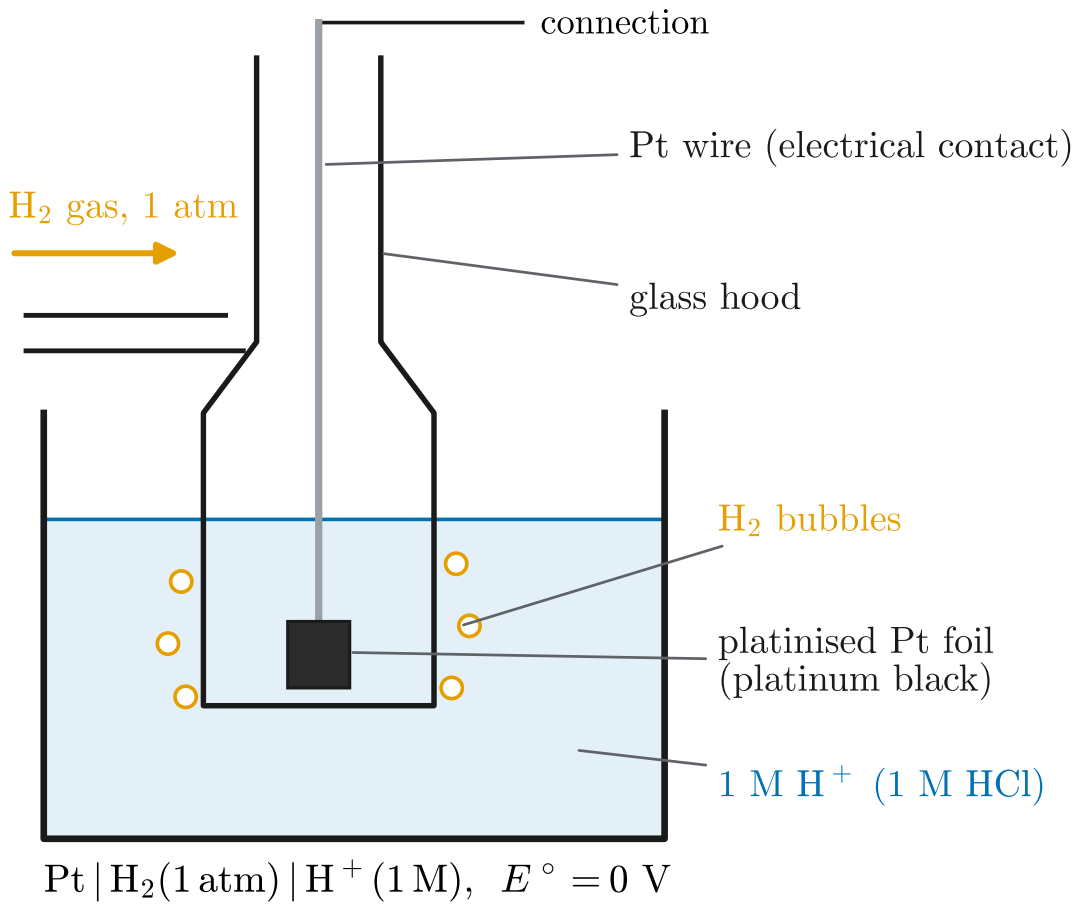


Figure G4.1. Standard hydrogen electrode (a gas electrode). ESA ★★

S003·5 · T003 00:05:49

How to read it. Hydrogen at 1 atm enters a glass hood and bubbles over a platinised Pt foil dipped in 1 M HCl; the Pt wire leads out to the connection. Label: $\mathbf{H_2}$ inlet, Pt wire, platinised foil, 1 M $\mathbf{H^+}$, bubbles. Redrawn from the deck's figure; schematic.

5 Amalgam electrode

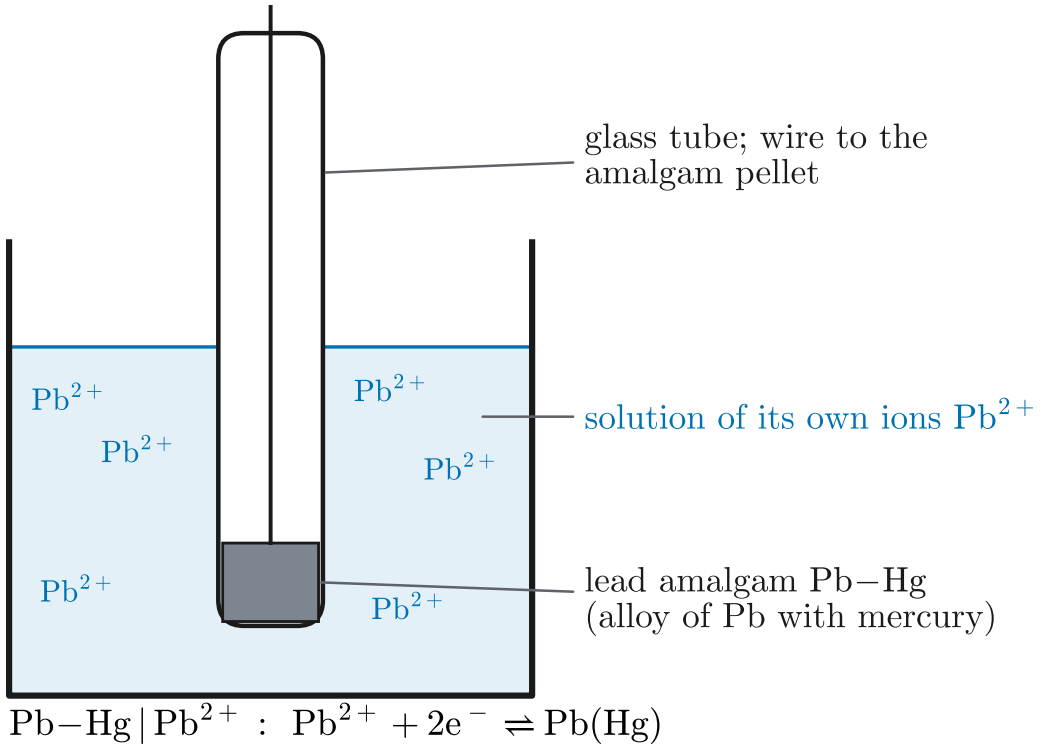


Figure G5.1. Lead amalgam electrode. ESA ★

S003·6

How to read it. A pellet of lead amalgam at the end of a glass tube (with a contact wire) dips in a solution of Pb^{2+}. It is a metal–metal ion electrode with the metal replaced by its amalgam. Simplified from the deck's figure.

6 Oxidation–reduction electrodes and the quinhydrone electrode

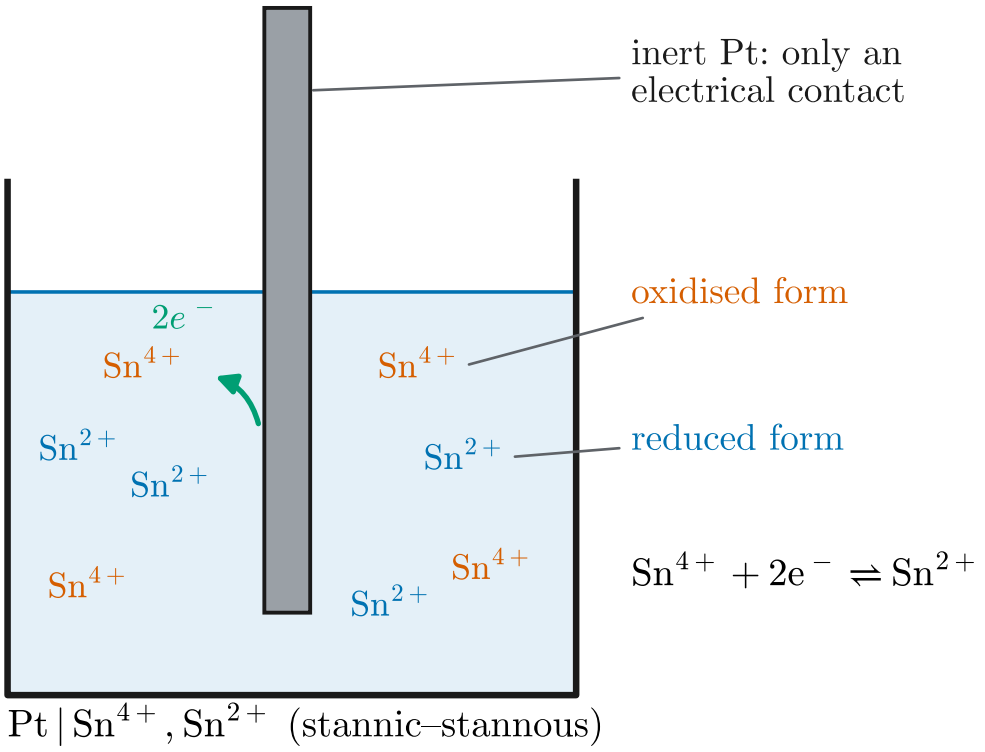


Figure G6.1. Redox electrode $\mathbf{Pt|Sn^{4+}, Sn^{2+}}$. ESA ★★

S003·7 · T003 00:12:19

How to read it. An inert Pt rod dips in a solution containing both $\mathbf{Sn^{4+}}$ (oxidised, vermilion) and $\mathbf{Sn^{2+}}$ (reduced, blue). Electrons exchanged between the two forms pass through the Pt (green arrow); the Pt itself does not react.

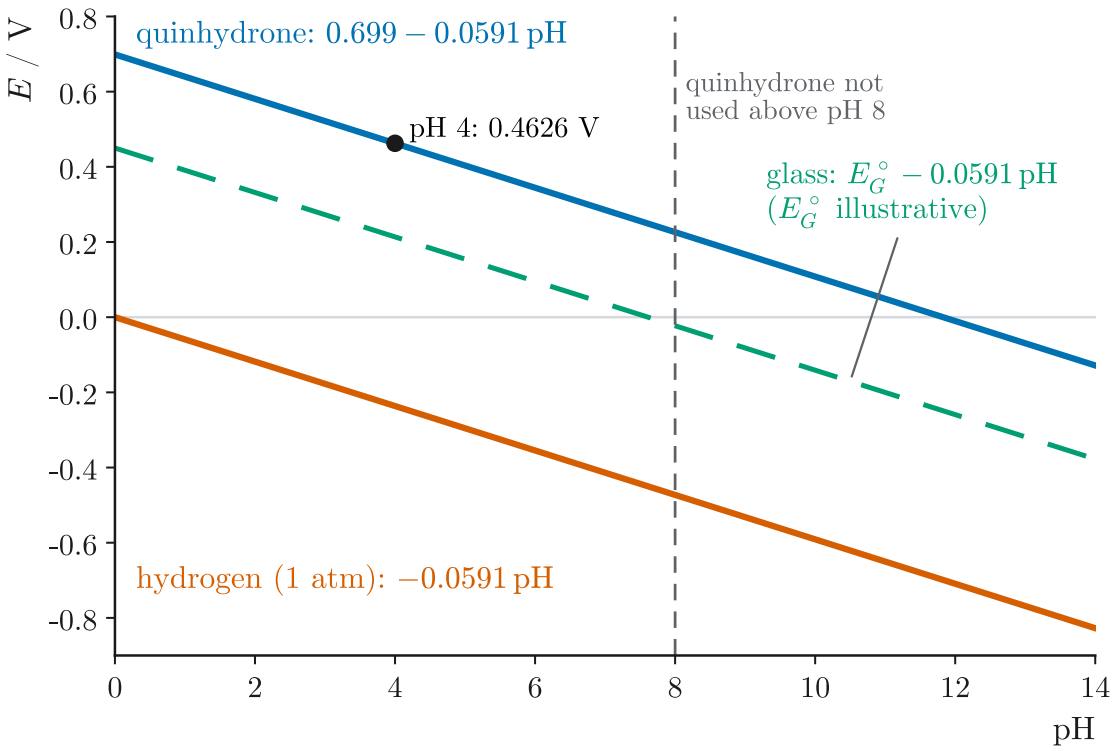


Figure G6.2. Potential against pH: hydrogen, quinhydrone and glass electrodes. ESA ★★

ADDED FOR CLARITY S003·5 · S003·8 · S003·9

How to read it. All three lines have slope −0.0591 V per pH unit at 298 K; they differ only in the intercept: 0 V (hydrogen, 1 atm), 0.699 V (quinhydrone), and E_G° (glass, illustrative value here). The dot is W6.3. The grey line marks the pH-8 limit of the quinhydrone electrode.

7 Ion-selective (membrane) electrode

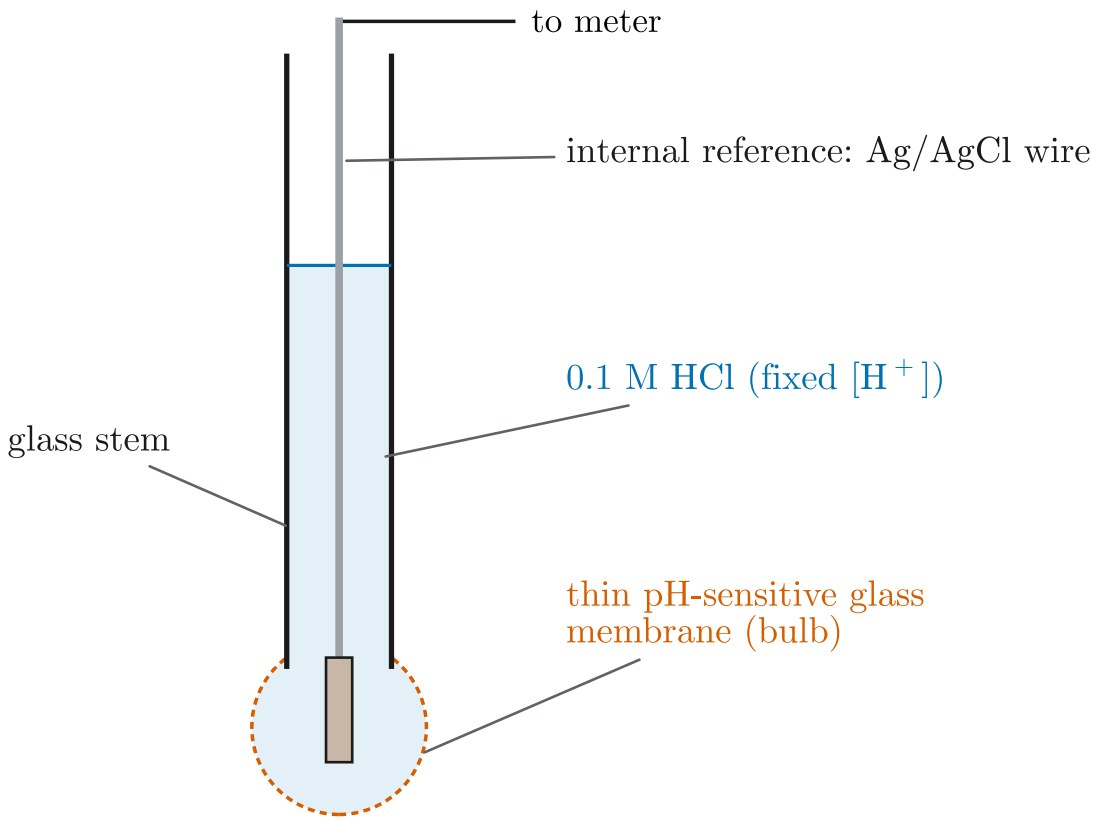


Figure G7.1. Glass electrode (simplified). ESA ★★

S003·9 · PDF p.56

How to read it. A glass tube ends in a thin pH-sensitive glass bulb (the membrane). Inside: 0.1 M HCl and an Ag/AgCl wire as the internal reference. The bulb dips in the test solution and is paired with an external reference such as the SCE. Construction details are taught in Unit 3.